Introduction to Derivatives

Assignment 1

1) The current level of the S&P 500 is $1040. The risk-free interest rate per year is 2%. Assume negligible dividends. The 6-month futures contract is trading at $1060.
   a) Is there an arbitrage opportunity? Explain.
   b) If there is an arbitrage opportunity, what strategy would you use to exploit it, without using any funds of your own? 2.5 points

2) The price of gold is currently $1,000 per ounce. The forward price for delivery in 1 year is $1,200. An arbitrageur can borrow money at 10% per annum. What should the arbitrageur do? Assume that the cost of storing gold is zero and that gold provides no income. 2.5 points

3) A forward contract on an asset is written at time zero. There are M periods until delivery. Say the carrying charge in period k is q*S(k), where S(k) is the spot price of the asset in period k. Also, say that d(0,M) is the discount factor from end of the last time period to time zero. Consider a portfolio that pays all carrying costs by selling a fraction of the asset as required.

   What is the Forward price? 5 points

Solutions

1)

$$1040(1+2\%)^{0.5} = 1050.35 \neq 1060$$

There is an arbitrage opportunity because the trading price is not equal to the "fair price" of the future. Borrow $ today, buy 1 unit of stock and short the future. You will make a riskless profit in 6m.

Security	Position	CF at 0	CF at 6 months
S&P 500 stock trading at $1,040	Buy 1	-1,040	S_T
Borrow / Lend Money	Borrow $1,040	1,040	-1,050.35
6 month future trading at $1,060	Short 1	0	$1,060 - S_T$
	TOTAL:	$0	$9.65

2) The arbitrageur should borrow money to buy a certain number of ounces of gold today and short forward contracts on the same number of ounces of gold for delivery in one year. This

means that gold is purchased for $1,000 per ounce and sold for $1,200 per ounce. Interest on the borrowed funds will be 0.1×$1,000 or $100 per ounce. A profit of $100 per ounce will therefore be made.

3)

(Proportional carrying charges) Suppose at time zero you take out a loan for $S(0)$, purchase 1 unit of the commodity, and short $(1-q)^M$ units of the commodity at a forward price of F per unit. The total intial cash outlay of these transactions is zero. Each period you pay the carrying cost of the commodity by selling a fraction q of your commodity holdings. Hence the amount of commodity held at the end of M periods is $(1-q)^M$. At the final time you deliver your commodity holdings to make good on your short, receiving $F(1-q)^M$. You also repay your loan by paying $S(0)/d(0,M)$. The total profit from these transactions (which clear all accounts) is $F(1-q)^M - S(0)/d(0,M)$. To avoid arbitrage, this profit must be zero. Hence $F = S(0)(1-q)^{-M}/d(0,M)$.
